Name
Date
Course

Limits (Homework)

Use the graph below to find the values of the expressions.

① $\lim_{x \to 3} f(x)$

② $f(3)$

③ $\lim_{x \to 1} f(x)$

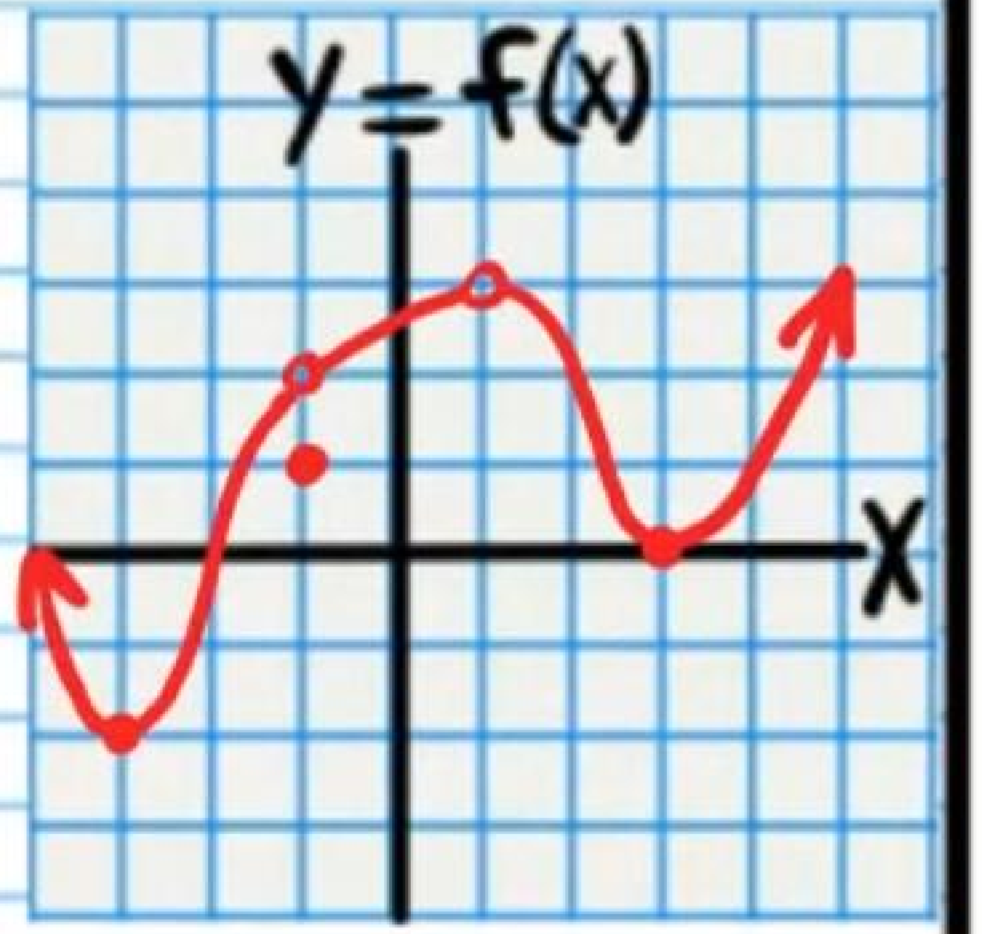
④ $f(1)$

⑤ $\lim_{x \to -1} f(x)$

⑥ $f(-1)$

⑦ $\lim_{x \to -3} f(x)$

⑧ $\lim_{x \to 2} f(x)$



Use the graph below to find the values of the expressions.

⑨ $\lim_{x \to 0} f(x)$

⑩ $\lim_{x \to -3} f(x)$

⑪ $\lim_{x \to 1} f(x)$

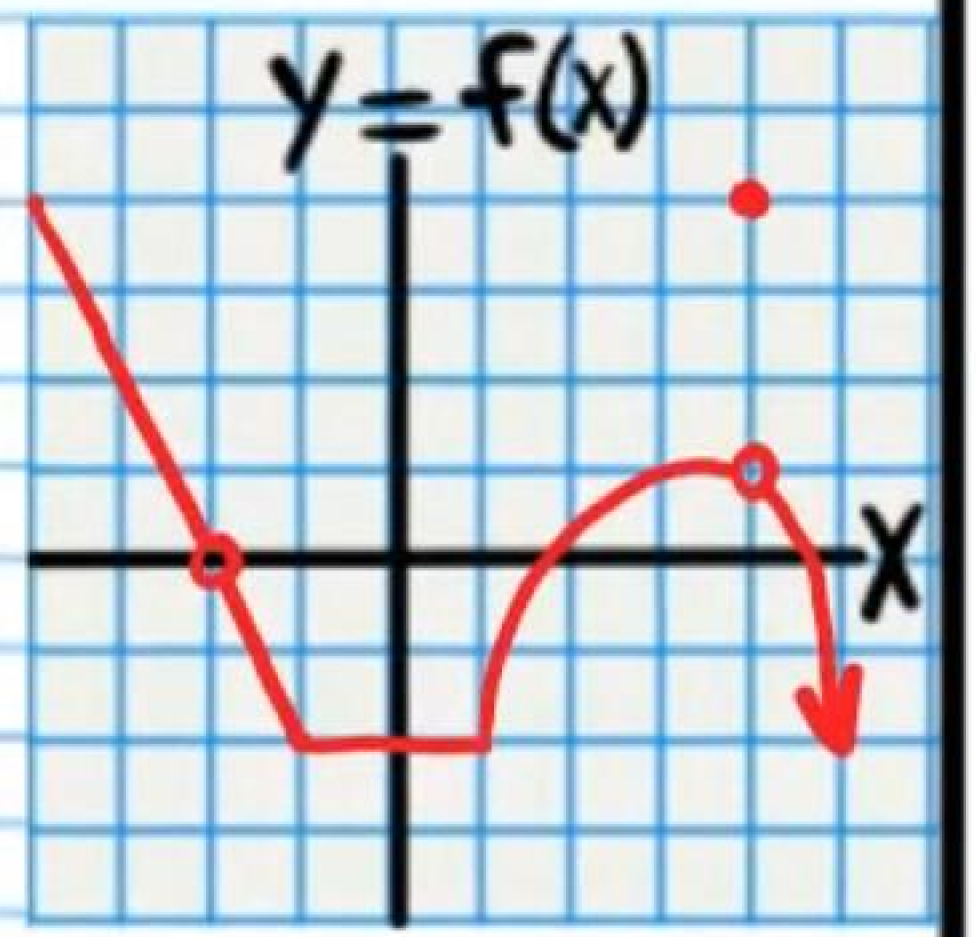
⑫ $f(1)$

⑬ $\lim_{x \to 4} f(x)$

⑭ $f(4)$

⑮ $\lim_{x \to -2} f(x)$

⑯ $f(-2)$



Use the graph below to find the values of the expressions.

⑰ $\lim_{x \to 2} f(x)$

⑱ $\lim_{x \to -1} f(x)$

⑲ $f(-1)$

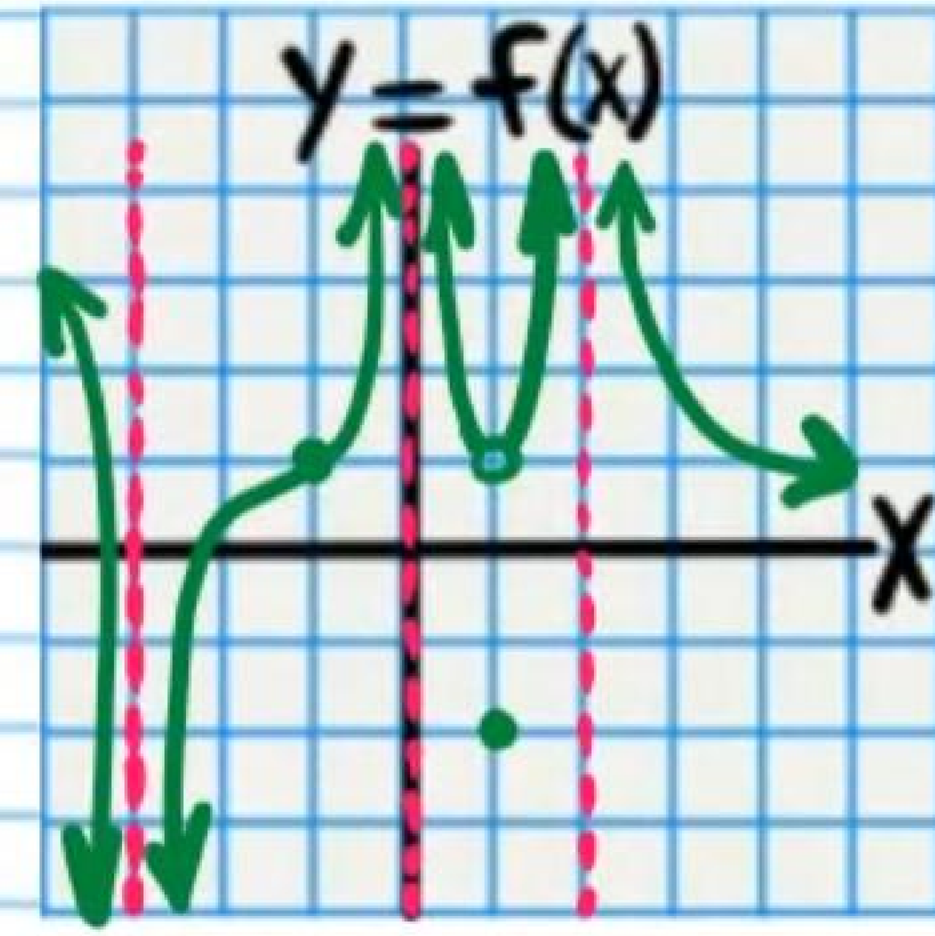
⑳ $\lim_{x \to 0} f(x)$

㉑ $\lim_{x \to 1} f(x)$

㉒ $f(1)$

㉓ $\lim_{x \to -3} f(x)$

㉔ $f(-3)$



Use the graph below to find the values of the expressions.

㉕ $\lim_{x \to 1} f(x)$

㉖ $\lim_{x \to 3} f(x)$

㉗ $f(3)$

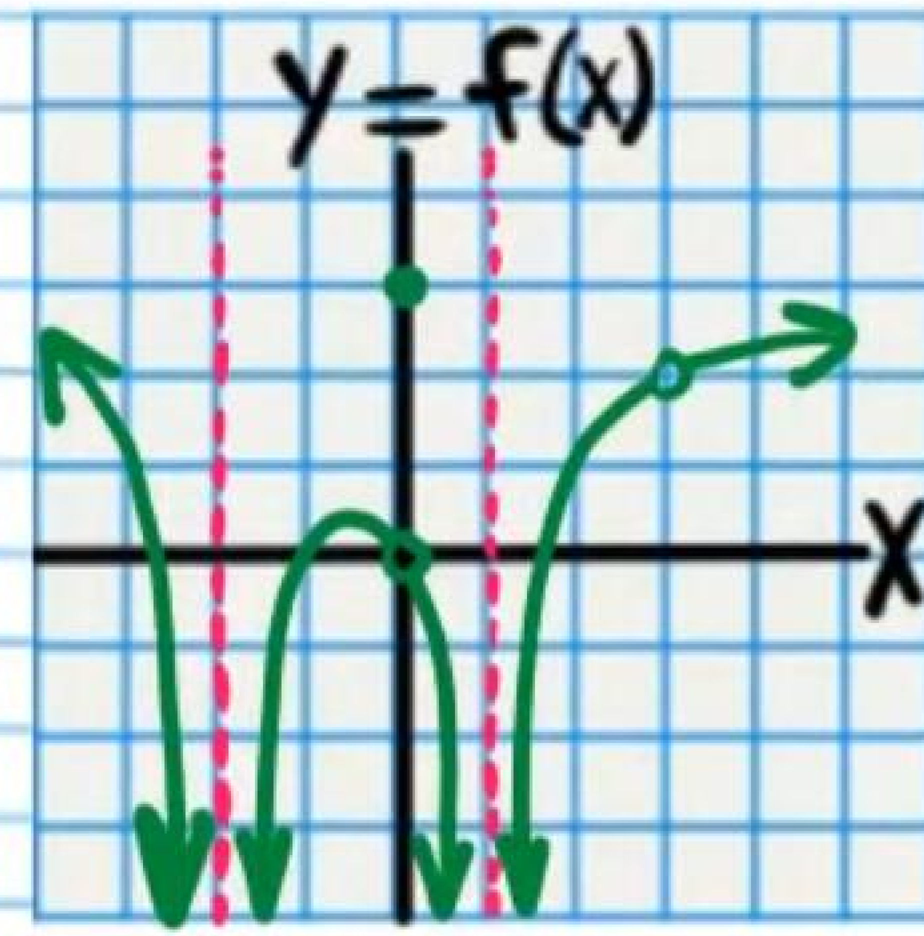
㉘ $\lim_{x \to -3} f(x)$

㉙ $\lim_{x \to 0} f(x)$

㉚ $f(0)$

㉛ $\lim_{x \to -2} f(x)$

㉜ $f(-2)$



Use the graph below to find the values of the expressions.

(33) $\lim_{x \to -2} f(x)$

(34) $f(-2)$

(35) $\lim_{x \to 0^-} f(x)$

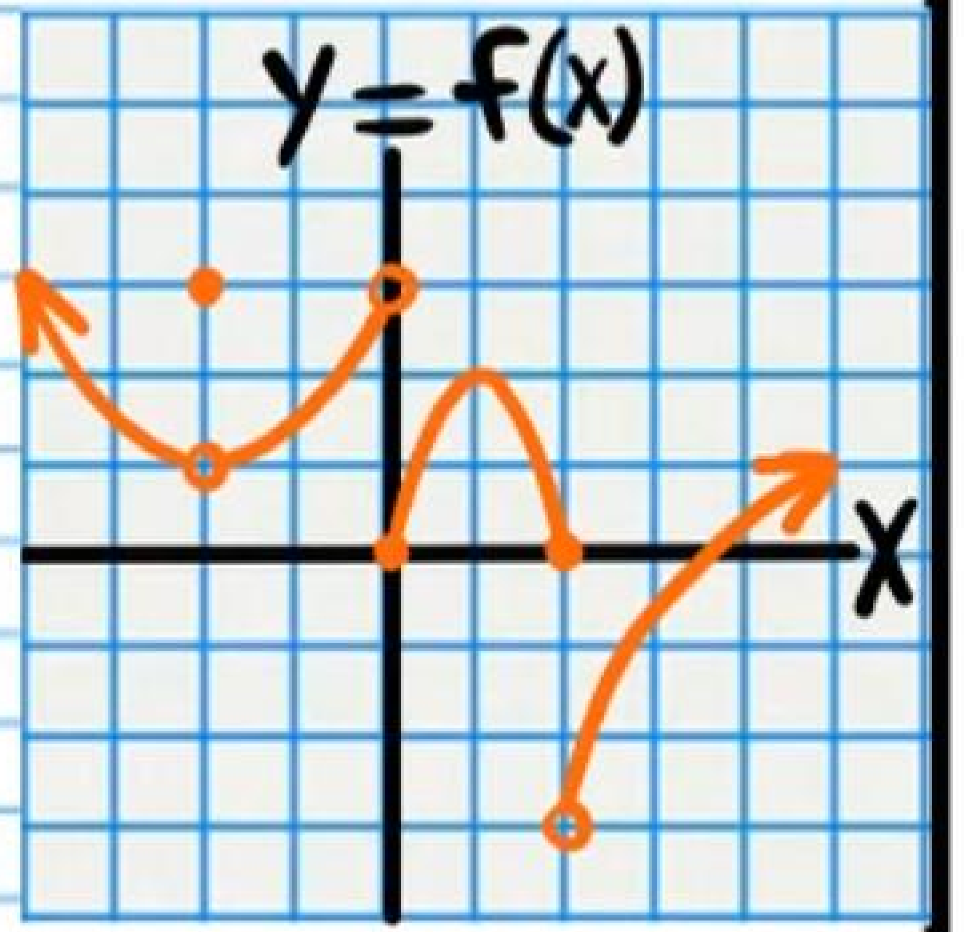
(36) $\lim_{x \to 0^+} f(x)$

(37) $\lim_{x \to 0} f(x)$

(38) $\lim_{x \to 2^-} f(x)$

(39) $\lim_{x \to 2^+} f(x)$

(40) $\lim_{x \to 2} f(x)$



Use the graph below to find the values of the expressions.

(41) $\lim_{x \to -1^-} f(x)$

(42) $\lim_{x \to -1^+} f(x)$

(43) $\lim_{x \to -1} f(x)$

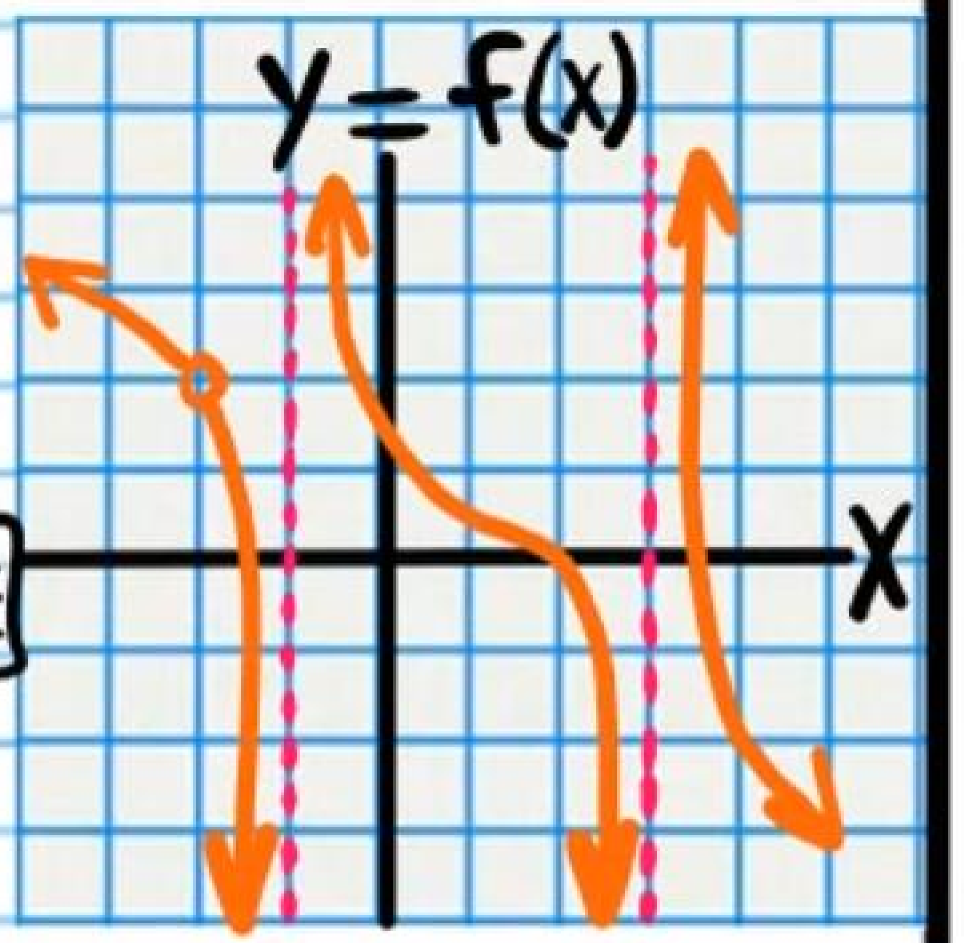
(44) $f(-1)$

(45) $\lim_{x \to 3^+} f(x) = \boxed{-3}$

(46) $\lim_{x \to 3^-} f(x) = \boxed{-1}$

(47) $\lim_{x \to 3} f(x) = \boxed{\text{DNE}}$

(48) $f(-2)$



Use the graph below to find the values of the expressions.

(49) $\lim_{x \to 2^-} f(x)$

(50) $\lim_{x \to 2^+} f(x)$

(51) $\lim_{x \to 2} f(x)$

(52) $f(2)$

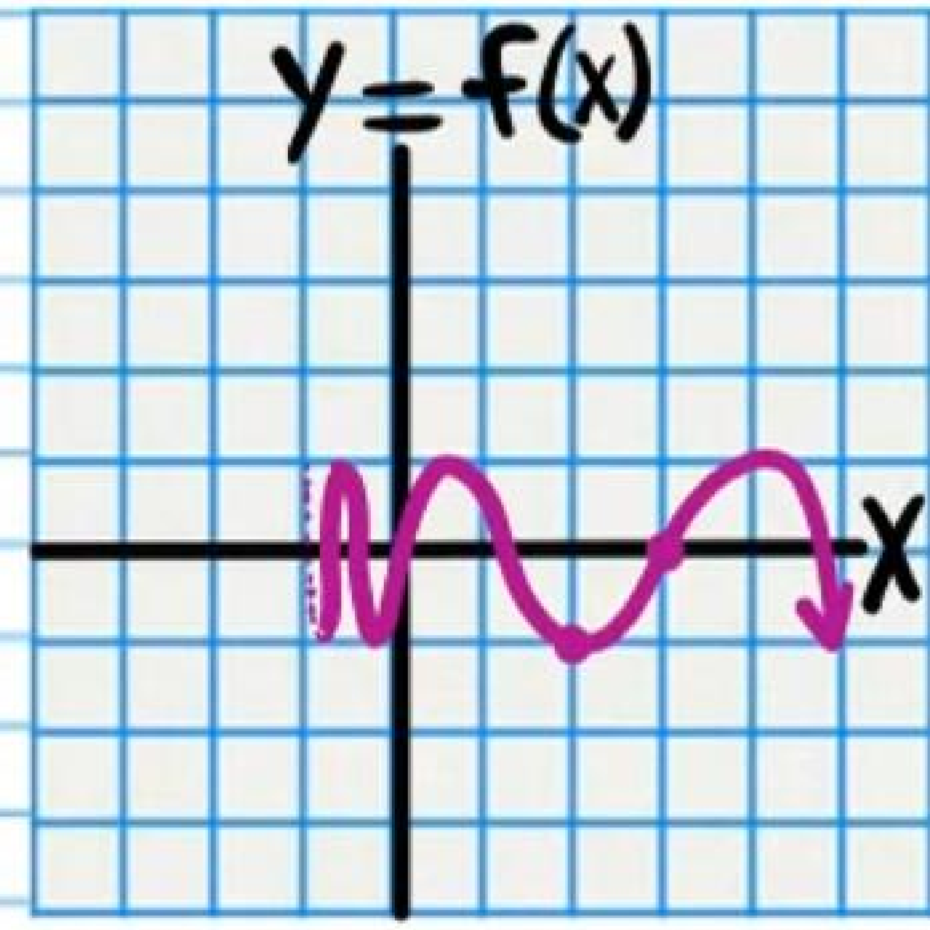
(53) $\lim_{x \to 3^-} f(x)$

(54) $\lim_{x \to 3^+} f(x)$

(55) $\lim_{x \to 3} f(x)$

(56) $f(3)$

(57) $\lim_{x \to -1^+} f(x)$



Find the value of each expression below.

(58) $\lim_{x \to 2} \sqrt[3]{4x}$

(59) $\lim_{x \to -9} (|x+2|-1)$

You may leave your answer in logarithm notation.

(60) $\lim_{x \to 3} \log_5\left(\frac{x}{2}\right)$

(61) $\lim_{x \to -1} \frac{5}{8x-2}$

(62) $\lim_{x \to \frac{\pi}{2}} \sin x$

(63) $\lim_{x \to 2} \frac{x-2}{x^2-7x+10}$

(64) $\lim_{x \to 0} \frac{2x^2-x}{x}$

(65) $\lim_{x \to -\frac{1}{7}} \frac{49x^2-1}{7x+1}$

(66) $\lim_{x \to 0} \frac{1-(x-1)^2}{x}$

67 $\lim_{x \to -5} \frac{x+5}{(x+3)^2-4}$

68 $\lim_{x \to 6} \frac{(x-7)^2-85+14x}{x-6}$

Homework Answers

1. 0
2. 0
3. 3
4. DNE
5. 2
6. 1
7. -2
8. 2
9. -2
10. 2
11. -2
12. -2
13. 1
14. 4
15. 0
16. DNE
17. ∞
18. 1
19. 1
20. ∞
21. 1
22. -2
23. $-\infty$
24. DNE
25. $-\infty$
26. 2
27. DNE
28. 1
29. 0
30. 3
31. $-\infty$
32. DNE
33. 1
34. 3
35. 3
36. 0
37. DNE
38. 0
39. -3
40. DNE
41. $-\infty$
42. ∞
43. DNE
44. DNE
45. ∞
46. $-\infty$
47. DNE
48. DNE
49. -1
50. -1
51. -1
52. -1
53. 0
54. 0
55. 0
56. 0
57. DNE
58. 2
59. 6
60. $\log_5\left(\frac{3}{2}\right)$
61. $-\frac{1}{2}$
62. 1
63. $-\frac{1}{3}$
64. -1
65. -2
66. 2
67. $-\frac{1}{4}$
68. 12